Test Series No.

1.7

MASTERY TEST SERIES

DURATION: 180 Minutes | **M. MARKS: 720**

DATE: 26/02/2026

SYLLABUS

Physics:	12th Full Syllabus
Chemistry:	12th Full Syllabus
Botany:	12th Full Syllabus
Zoology:	12th Full Syllabus

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. Let the charge density distribution of a solid sphere of radius R and total charge Q be given by $\rho(r)=\frac{Q}{\pi R^4}r$. For a point P inside the sphere at distance r_1 from the centre of the sphere, the magnitude of electric field is:

(1) Zero (2) $\frac{Q}{4\pi\epsilon_0 r_1^2}$

(3) $\frac{Qr_1^2}{4\pi\epsilon_0 R^4}$ (4) $\frac{Qr_1^2}{3\pi\epsilon_0 R^4}$

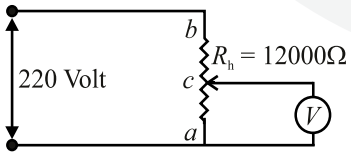
2. A free neutron decays into a proton but a free proton does not decay into neutron. This is because

(1) neutron is an uncharged particle
(2) proton is a charged particle
(3) neutron is a composite particle made of a proton and an electron
(4) neutron has larger rest mass than proton

3. A light wave traveling from air through a material where wave has a frequency of 4×10^{14} Hz and a wavelength of 5×10^{-7} m. The refractive index of the material is:

(1) 1.33 (2) 2.0
(3) 1.5 (4) 1.2

4. A potential difference of 220V is maintained across a 12000 Ω rheostat as shown in fig. The voltmeter V has resistance of 6000 Ω and the point C is at one fourth of the distance from a to b. The reading of voltmeter will be:

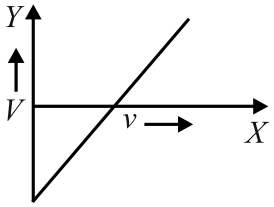


(1) 30V (2) 40V
(3) 50V (4) 60V

5. A capacitor with capacitance 20 pF is connected to a circuit where the potential difference across its plates changes at a rate of 2×10^{11} V/s. What is the displacement current?

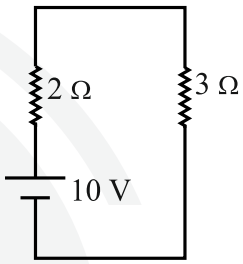
(1) 0.4 A (2) 2 A
(3) 4 A (4) 10 A

6. The stopping potential V for photo-electric emission from a metal surface is plotted along Y-axis and frequency v of incident light along X-axis. A straight line is obtained as shown. Planck's constant is given by



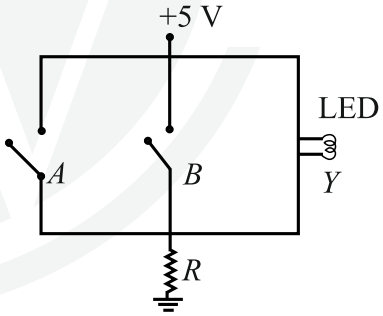
(1) Slope of the line
(2) Product of slope of the line and magnitude of charge on the electron
(3) Product of intercept along Y-axis and mass of the electron
(4) Product of slope and mass of electron

7. In the circuit shown, a 10V battery is connected to two resistors of 2 Ω and 3 Ω. The power delivered by the battery is:



(1) 10 W (2) 20 W
(3) 25 W (4) 50 W

8. Name the logic gate equivalent to the diagram attached:



(1) OR (2) NOR
(3) NAND (4) AND

9. An LCR series circuit consists of an inductance $L = 50$ mH, a capacitance $C = 20\mu$F, and a resistance $R = 40\ \Omega$. At what angular frequency will the current amplitude be maximum?

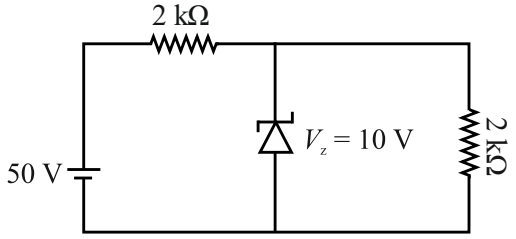
(1) 500 rad/s (2) 1000 rad/s
(3) 2500 rad/s (4) 100 rad/s

10. Light of wavelength 6328 Å is incident normally on a slit having a width of 0.2 mm. The angular width of the central maximum measured from minimum to minimum of diffraction pattern on a screen 9 m away will be nearly

(1) 0.36 degree (2) 0.18 degree
(3) 0.72 degree (4) 0.09 degree

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11. In the given Zener diode circuit, $V_{in} = 50$ V, series resistance $R = 2\ \text{k}\Omega$, and the Zener breakdown voltage is $V_z = 10$ V. If the load resistance is $2\ \text{k}\Omega$, the current flowing through the Zener diode is:



(1) 5 mA (2) 10 mA
(3) 20 mA (4) 15 mA

12. If in a plano-convex lens, radius of curvature of convex surface is 10 cm and the focal length of lens is 30 cm, the refractive index of the material of the lens will be
(1) 1.5 (2) 1.66
(3) 1.33 (4) 3

13. A proton is moving with a constant velocity along the +y axis. At the instant it is at origin, what is the direction of the magnetic field produced by this proton at a point on the +z axis?
(1) +x axis (2) -x axis
(3) +y axis (4) +z axis

14. Angular width of central maximum of a diffraction pattern on a single slit does not depend upon
(1) Distance between slit and source
(2) Wavelength of the light used
(3) Width of the slit
(4) Frequency of light used

15. An LED is fabricated from a semiconductor with an energy gap of 2.48 eV. Determine the wavelength and color of the emitted light. (Take $hc \approx 1240$ eV-nm).
(1) 500 nm, Green (2) 620 nm, Red
(3) 450 nm, Blue (4) 580 nm, Yellow

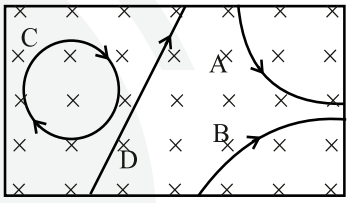
16. Choose the option containing **correct** statement.
(1) The terminal potential difference across a cell is always less than its emf.
(2) The terminal potential difference across a cell is always equal to its emf.
(3) The terminal potential difference across a cell is always greater than its emf.
(4) The terminal potential difference across a cell may be less than or greater than emf of the cell.

17. A material placed in a magnetizing field of intensity 2000 A/m acquires a magnetic flux density of 0.6 T. The magnetic permeability of the material is:
(1) 3×10^{-4} T · m/A
(2) 1.2×10^{-3} T · m/A
(3) 3×10^{-5} T · m/A
(4) 6×10^{-4} T · m/A

18. Points A and B are situated perpendicular to the axis of a short bar magnet at large distances x and 3x from its centre on opposite sides. The ratio of the magnitude of magnetic fields at A and B will be approximately equal to:
(1) 2 : 9 (2) 1 : 9
(3) 27 : 1 (4) 9 : 1

19. A nuclide X absorbs a neutron to form an intermediate Y, which immediately decays into an α -particle and ${}^{11}_{5}$B. Identify the nuclide X. (Reaction: ${}^{A}_{Z}X + {}^{1}_{0}n \rightarrow Y \rightarrow {}^{4}_{2}He + {}^{11}_{5}B$)
(1) ${}^{15}_{7}$N (2) ${}^{14}_{7}$N
(3) ${}^{14}_{6}$C (4) ${}^{16}_{8}$O

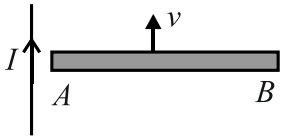
20. Four particles enter a uniform magnetic field directed into the page. Based on their trajectories, consider these statements:



I. Particle A is positively charged.
II. Particle B is negatively charged.
III. Particle C is negatively charged.
IV. Particle D is positively charged.
Which of the above statement(s) are **correct**?
(1) I, II, and III only
(2) II, III, and IV only
(3) I, III, and IV only
(4) All are correct

21. In the β^- decay process described by ${}^{14}_{6}C \rightarrow {}^{14}_{7}N + e^- + X$, the particle X is:
(1) Proton (2) Neutron
(3) Antineutrino (4) Neutrino

22. The current carrying wire and the rod AB are in same plane. The rod moves parallel to the wire with a velocity v. Which one of the following statement is true about induced emf in the rod?



(1) End A will be at lower potential with respect to B
(2) A and B will be at same potential
(3) There will be no induced emf in the rod
(4) Potential at A will be higher than that at B

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23. According to the Bohr quantization condition and de-Broglie hypothesis, which of the following expressions represents a valid radius (r) for a stable electron orbit? (λ is the de-Broglie wavelength)

(1) $r=\frac{\lambda}{4\pi}$ (2) $r=\frac{3\lambda}{2\pi}$

(3) $r=\frac{4\lambda}{3\pi}$ (4) $r=\frac{5\lambda}{4\pi}$

24. Select the **incorrect** statement

(1) In a diamagnetic substance, net magnetic moment of each atom/molecule is zero

(2) In a paramagnetic substance, net magnetic moment of each atom/molecule is non-zero

(3) In a ferromagnetic material, net magnetic moment of each domain is zero

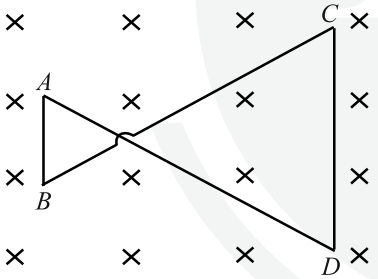
(4) Both (1) and (2)

25. In Bohr's model of the hydrogen atom, what is the ratio of the time period of revolution of an electron in orbit $n=2$ to that in orbit $n=3$?

(1) 8: 27 (2) 4: 9

(3) 1: 8 (4) 27: 8

26. A conducting wire frame is placed in magnetic field which is directed into the paper. The magnetic field is increasing at a constant rate. The directions of induced current in wires AB and CD are



(1) B to A and D to C (2) A to B and C to D

(3) A to B and D to C (4) B to A and C to D

27. If the wavelength of the shortest wavelength of the Lyman series is λ_L, what is the shortest wavelength of the Balmer series?

(1) λ_L (2) $2\lambda_L$

(3) $16\lambda_L$ (4) $4\lambda_L$

28. Which of the following statement is false for the properties of electromagnetic waves?

(1) These waves do not require any material medium for propagation

(2) Both electric and magnetic field vectors oscillate in phase with each other

(3) The energy in electromagnetic wave is divided equally between electric and magnetic field vectors

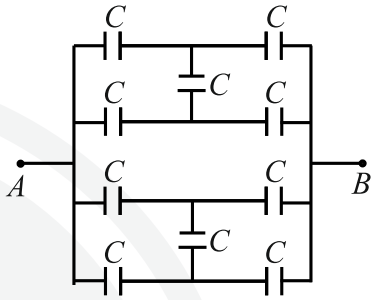
(4) Both electric and magnetic field vectors are parallel to each other and perpendicular to the direction of propagation of wave

29. In an electron microscope, electrons are accelerated through accelerating voltage of 10 kV, they have de-brogle wavelength λ. If the voltage is increased to 40 kV, how does the new de-Broglie wavelength (λ') compare to the initial wavelength (λ)?

(1) $\lambda'=2\lambda$ (2) $\lambda'=4\lambda$

(3) $\lambda'=\frac{\lambda}{2}$ (4) $\lambda'=\frac{\lambda}{4}$

30. The effective capacitance of combination of equal capacitors each having capacitance C between points A and B as shown in figure is:



(1) C (2) 2C

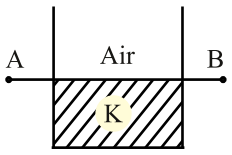
(3) 3C (4) 4C

31. When a metal surface is illuminated with light of wavelength λ, the stopping potential is V. When the same surface is illuminated with wavelength 3λ, the stopping potential drops to $\frac{V}{4}$. The threshold wavelength for this metal is:

(1) 5λ (2) 3λ

(3) 4λ (4) 9λ

32. A parallel plate air capacitor has capacitance C across A and B. Now half of the space is filled with a dielectric material with dielectric constant K as shown in the figure. The new capacitance is C' then:



(1) $C'=\frac{C}{2}[1+K]$ (2) $C'=C[1+\mathrm{K}]$

(3) $C'=C\left[1+\frac{K}{2}\right]$ (4) $C'=C\left[\frac{1}{2}+K\right]$

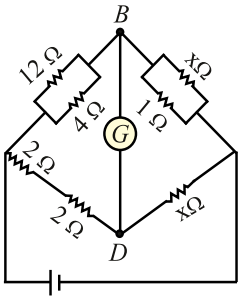
33. Unpolarized light falls on a system of two polaroids. If the pass axis of the second polaroid is at 45° to the pass axis of the first, what percentage of the incident light intensity is transmitted?

(1) 25% (2) 50%

(3) 75% (4) 12.5%

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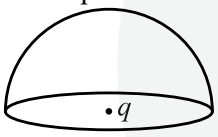
34. In the following Wheatstone bridge circuit, the potential difference between points *B* and *D* is zero. If the value of x is $\frac{1}{n}\Omega$, find the value of n.



(1) n = 3 (2) n = 2
(3) n = 4 (4) n = 1

35. A glass prism in air has a refracting angle of 60º and a refractive index of $\sqrt{3}$. Calculate the angle of incidence required for the ray to undergo minimum deviation.
(1) 30º (2) 60º
(3) 45º (4) 90º

36. A point charge $q = 17.7\times10^{-12}$ C is placed at the base center of a hemisphere (as shown in the figure). What is the electric flux passing through the curved surface of the hemisphere?



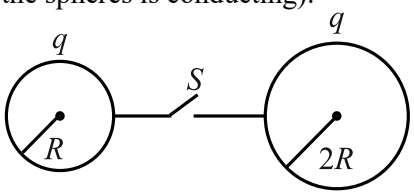
(Take $\epsilon_0 = 8.85\times10^{-12}$ $C^2 N^{-1} m^{-2}$)
(1) 2.0 N m^2/C (2) 0.5 N m^2/C
(3) 1.0 N m^2/C (4) 4.0 N m^2/C

37. The magnetic flux linkage of a coil of resistance 10 Ω changes from 10 Wb to 50 Wb in the same direction in 2 s. The total electric charge that flows through the coil is:
(1) 2 C (2) 8 C
(3) 20 C (4) 4 C

38. The specific charge (charge to mass ratio) of an electron is approximately 1.76×10^{11} C/kg. What is the total magnitude of charge possessed by 1 mg of electrons?
(1) 1.76×10^5 C (2) 1.76×10^8 C
(3) 1.76×10^4 C (4) 1.76×10^7 C

39. An AC circuit has an instantaneous voltage $V = 100\sin(100t)$ volts and instantaneous current $I = 4\sin\left(100t + \frac{\pi}{3}\right)$ A. What is the impedance of the circuit?
(1) 400 Ω (2) 25 Ω
(3) 50 Ω (4) 12.5 Ω

40. Two conducting spheres, each given a charge *q*, are kept far apart as shown. The amount of charge that crosses the switch *S*, when it is closed, until equilibrium is reached, is (the connection between the spheres is conducting):



(1) $\frac{q}{3}$ (2) $\frac{2q}{3}$
(3) $\frac{3q}{4}$ (4) Zero

41. A non-conducting solid sphere of radius R is uniformly charged with a total charge Q. At what distance from the center of the sphere is the electric potential equal to one-third of the potential at the center?
(1) R (2) 3R
(3) 2R (4) 1.5R

42. Suppose net magnetic force acting on a moving point charge *q* at certain instant in a region of uniform magnetic field $\vec{B}$ is non-zero. Angle between its velocity $\vec{v}$ and magnetic field at that instant:
(1) must be $\frac{\pi}{2}$
(2) may be all angles except $\frac{\pi}{2}$
(3) must be zero or π
(4) can be any angle except zero or π

43. In a plane electromagnetic wave in vacuum, the amplitude of the electric field is 60 V/m. The amplitude of the oscillating magnetic field is:
(1) 2×10^{-7} T (2) 6×10^{-7} T
(3) 1.8×10^{-7} T (4) 3×10^{-8} T

44. In India electricity is supplied for domestic use at 220 V. It is supplied at 110 V in USA. If the resistance of a 60 W bulb for use in India is *R*, the resistance of a 60 W bulb for use in USA will be:
(1) *R* (2) 2*R*
(3) $\frac{R}{4}$ (4) $\frac{R}{2}$

45. A wire of length *L* is cut into two parts of lengths *A* and *B*. Part *A* is stretched to double its original length. If the resistance of the stretched wire *A* becomes equal to the resistance of the wire *B*, what was the ratio of the original lengths of *A* and *B* $(L_A : L_B)$?
(1) 1: 2 (2) 1: 4
(3) 1: 1 (4) 1: 8

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46. Which of the following carbonyl compound will undergo cannizzaro reaction?
(1) PhCHO
(2) CH_3CHO
(3) PhCOPh
(4) $PhCH_2CHO$

47. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: $[Mn_2(CO)_{10}]$ is an organometallic compound.
Reason R: It involves synergic bonding.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

48. Which among the following is an example of pseudo first order reaction?
(1) Hydrolysis of ethyl acetate in basic medium
(2) Decomposition of N_2O
(3) Hydrogenation of ethene
(4) Hydrolysis of ethyl acetate in acidic medium

49. The reagent neutral ferric chloride is used for the detection of:
A. Sulphide ion
B. Alcoholic –OH group
C. Acetate ion
(1) A only
(2) B only
(3) C only
(4) A, B and C

50. The strongest reducing agent among the following is:
(1) Al(s)
(2) Fe s)
(3) Cu(s)
(4) Ag(s)

51. The **correct** order of acidic strength for the given hydrides is:
(1) $H_2S < H_2Se < H_2Te < H_2O$
(2) $H_2S < H_2O < H_2Se < H_2Te$
(3) $H_2Se < H_2S < H_2O < H_2Te$
(4) $H_2O < H_2S < H_2Se < H_2Te$

52. Spin only magnetic moment of the complex ion, $[Ni(H_2O)_4]^{2+}$ is:
(1) zero BM (2) $\sqrt{3}$ BM
(3) $\sqrt{24}$ BM (4) $\sqrt{8}$ BM

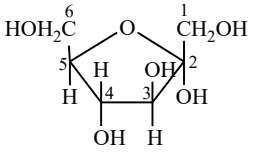
53. Which of the following metals exhibit only +3 oxidation state?
(1) Sc (2) V
(3) Ti (4) Cr

54. 0.1 M aqueous solution of which of the following solutes will has highest osmotic pressure at 27°C?
(1) NaCl (2) $K_4[Fe(CN)_6]$
(3) $CaCl_2$ (4) Glucose

55. Given below are two statements:
Statement I: The number of resonating structures of aniline is six.
Statement II: Sulphanilic acid exist as zwitter ion.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

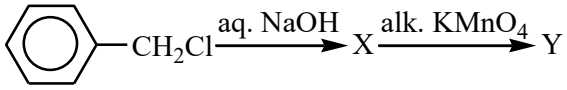
56. Essential amino acid among the following is:
(1) Alanine (2) Glutamine
(3) Histidine (4) Aspartic acid

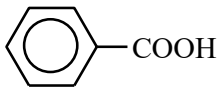
57. Major product of the following reaction is:
$Ph-CN \xrightarrow[(ii)\ H_2O]{(i)\ PhMgBr}$
(1) Quinol
(2) Cresol
(3) Benzophenone
(4) Benzoquinone

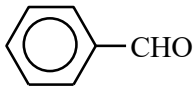
58.  represents:
(1) α-D-(–)-Fructofuranose
(2) β-D-(–)-Fructofuranose
(3) α-D-(+)-Fructofuranose
(4) β-D-(+)-Fructofuranose

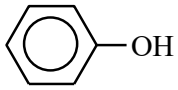
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59. The major product (Y) in the following reaction sequence is:



(1) 

(2) 

(3) 

(4) 

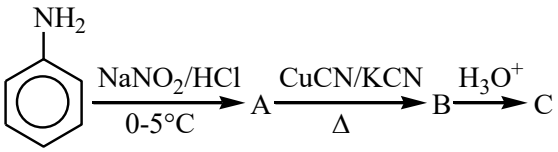
60. **Incorrect** statement regarding S_N2 reaction of alkyl halides is:
(1) proceed through via transition state
(2) Inversion in configuration take place
(3) one step reaction
(4) favoured in polar protic solvent.

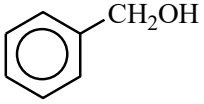
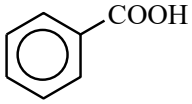
61. The **correct** order of negative electron gain enthalpy is:
(1) F > Cl > Br > I (2) Cl > F > Br > I
(3) Cl > Br > F > I (4) Cl > Br > I > F

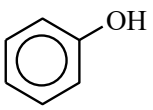
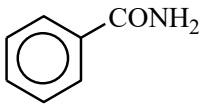
62. All the given compounds are complex salts **except**:
(1) $[Co(en)_3]Cl_3$
(2) $FeSO_4.(NH_4)_2SO_4.6H_2O$
(3) $NiCl_2.6H_2O$
(4) $PtCl_2.2NH_3$

63. The half life of a first order reaction which takes one minutes for 90% completion is:
[Given: log 5 = 0.7]
(1) 0.2 min (2) 0.3 min
(3) 0.4 min (4) 0.5 min

64. The major product (C) obtained in the given reaction sequence is:



(1)  (2) 

(3)  (4) 

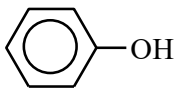
65. Quantity of electricity required to convert 0.1 mol of MnO_4^{2-} to MnO_4^- is:
(1) 9650 C
(2) 14475 C
(3) 7237.6 C
(4) 13027 C

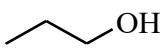
66. Zr and Hf are:
(1) Transuranic elements
(2) Representative elements
(3) f-block elements
(4) Twins elements

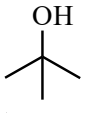
67. The percentage composition of C_2H_5OH in azeotrope of C_2H_5OH and H_2O by volume is:
(1) 85%
(2) 95%
(3) 75%
(4) 65%

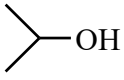
68. Given below are two statements:
Statement I: All B group vitamins can not be stored in our body.
Statement II: Lactose is a non-reducing sugar.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

69. The alcohol which react with fastest rate with conc. HCl and $ZnCl_2$ is:

(1) 

(2) 

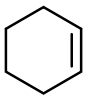
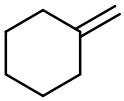
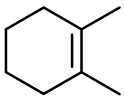
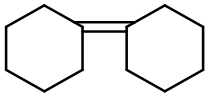
(3) 

(4) 

70. Number of structural isomers for molecular formula $C_4H_{11}N$ is:
(1) 5
(2) 6
(3) 7
(4) 8

[Contd...

71. First inert compound is formed by Neil bartlett by reacting xenon with:
(1) PtF_6
(2) Ag_2O
(3) F_2
(4) O_2

72. Kharash effect is observed in:
(1) 
(2) 
(3) 
(4) 

73. Consider the experimental observation for the reaction $A + B \rightarrow C$

A[M]	B[M]	rate[Ms^{-1}]
0.1	0.2	1.5×10^{-4}
0.2	0.2	1.5×10^{-4}
0.2	0.4	3×10^{-4}

The overall order of the reaction is:
(1) zero
(2) 1
(3) 1.5
(4) 2

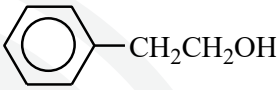
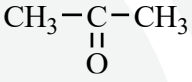
74. $NH_3(g)$ on passing through Nessler's reagent gives:
(1) Red precipitate
(2) Blue precipitate
(3) Brown precipitate
(4) Green precipitate

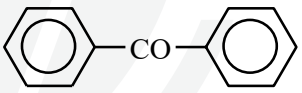
75. When an external opposite potential is applied to standard Daniell cell, the reaction continues to take place till the opposite voltage reaches the value of:
(1) 2.2V (2) 1.1V
(3) 0.2V (4) 0.1V

76. The **incorrect** statement regarding actinoids is:
(1) They show actinoid contraction due to poor shielding by 5f-electrons.
(2) These are highly reactive metals.
(3) Their generally oxidation state is +3.
(4) Their magnetic properties are more simple than lanthanoids.

77. Least soluble gas in water among the following at 25°C is:
(1) P (K_H = 2.31 kbar)
(2) Q (K_H = 0.33 kbar)
(3) R ($K_H = 2.41 \times 10^{-4}$ kbar)
(4) S (K_H = 20.2 kbar)

78. When ethanol is treated with H_2SO_4, the major products obtain at 413 K and 443 K respectively are:
(1) $(C_2H_5)_2O$ and $(C_2H_5)_2SO_4$
(2) C_2H_4 and $(C_2H_5)_2SO_4$
(3) $(C_2H_5)_2O$ and C_2H_4
(4) C_2H_4 and $(C_2H_5)_2O$

79. Which of the following compounds will give iodoform on reaction with I_2/KOH?
(1) $CH_3CH_2CH_2OH$
(2) 
(3) HCHO
(4) 

80. Most reactive compound towards nucleophilic addition reaction among the following is:
(1) CH_3CHO
(2) HCHO
(3) CH_3COCH_3
(4) 

81. Given below are two statements:
Statement I: Nitrogen is restricted to a maximum covalency of three.
Statement II: Among 15th group hydrides, NH_3 has maximum boiling point.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

82. C_2H_5F can be prepared by:
(1) Dow's process
(2) Swarts reaction
(3) Finkelstein reaction
(4) Sandmeyer's reaction

[Contd...

83. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: Phenol on reaction with CH_3MgBr gives benzene.
Reason R: Phenol has acidic H-atom.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

84. The mole fraction of toluene in vapour phase in equilibrium with 2:1 molar ratio of benzene and toluene solution at t°C is:
(Given: $p^0_{Benzene} = 80$ torr and $p^0_{Toluene} = 24$ torr)
(1) 0.13
(2) 0.21
(3) 0.36
(4) 0.07

85. Given below are two statements:
Statement I: $[Al(C_2O_4)_3]^{3-}$ is a homoleptic complex ion.
Statement II: Oxidation number of nickel in $[Ni(CO)_4]$ is zero.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

86. Match **List-I** with **List-II**.

List-I (Order of reaction)		**List-II (Unit of rate constant)**	
(A)	First	(I)	$mol\ L^{-1}\ s^{-1}$
(B)	Second	(II)	$mol^{-1}\ L\ s^{-1}$
(C)	Zero	(III)	s^{-1}
(D)	Third	(IV)	$mol^{-2}\ L^2\ s^{-1}$

Choose the **correct** answer from the options given below:
(1) A-III; B-I; C-IV; D-II
(2) A-III; B-II; C-I; D-IV
(3) A-II; B-IV; C-I; D-III
(4) A-I; B-III; C-IV; D-II

87. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: For weak electrolytes, specific conductance increases on dilution.
Reason R: On dilution degree of dissociation of weak electrolyte increases.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

88. Match **List-I** with **List-II**.

List-I (Aquated metal ions)		**List-II (Colour)**	
(A)	Zn^{2+}	(I)	Blue
(B)	Mn^{2+}	(II)	Colourless
(C)	Cu^{2+}	(III)	Pink
(D)	Fe^{2+}	(IV)	Green

Choose the **correct** answer from the options given below:
(1) A-III; B-I; C-IV; D-II
(2) A-II; B-III; C-I; D-IV
(3) A-II; B-IV; C-I; D-III
(4) A-I; B-III; C-IV; D-II

89. During denaturation of protein which structure remain intact?
(1) Quaternary structure
(2) Tertiary structure
(3) Secondary structure
(4) Primary structure

90. Match **List-I** with **List-II**.

List-I (Amine)		**List-II (pK_b value in aqueous phase)**	
(A)	Methanamine	(I)	3.27
(B)	N,N-dimethylaniline	(II)	3.38
(C)	N-methylmethanamine	(III)	4.7
(D)	Phenylmethanamine	(IV)	8.92

Choose the **correct** answer from the options given below:
(1) A-II; B-IV; C-I; D-III
(2) A-II; B-III; C-I; D-IV
(3) A-III; B-I; C-IV; D-II
(4) A-I; B-III; C-IV; D-II

[Contd...

91. Given below are two statements:
Statement I: Panchanan Maheshwari's work on test tube fertilisation and intra-ovarian pollination won worldwide acclaim.
Statement II: Panchanan Maheshwari emphasised the need for initiation of work on artificial culture of immature embryos.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

92. More than 70 per cent of all the species recorded on earth are;
(1) insects. (2) plants.
(3) animals. (4) fungi.

93. Read the following statements regarding decomposition:
A. Decomposition is largely an oxygen-requiring process.
B. Decomposition rate is slower if detritus is rich in nitrogen and water-soluble substances like sugars.
C. Low temperature and anaerobiosis inhibit decomposition.
D. Humification leads to accumulation of a dark coloured amorphous substance called humus.
E. Humus is highly resistant to microbial action.
The **correct** statements are;
(1) A, C, D and E
(2) A, B, C and D
(3) B, C, D and E
(4) A, B, D and E

94. A male honey bee;
(1) is diploid.
(2) is produced by fertilisation of gametes.
(3) can have father and sons.
(4) is also called a drone.

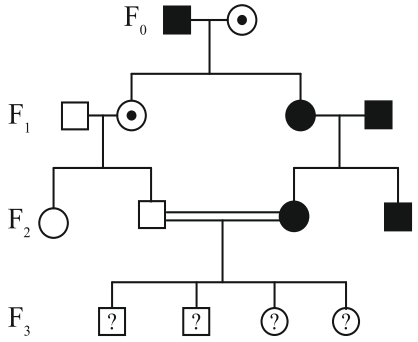
95. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): DNA is a better genetic material than RNA.
Reason (R): DNA is a polymer of deoxyribonucleotides.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

96. A population age pyramid with a broad base that narrows towards the apex reflects which growth status of the population?
(1) Stable
(2) Declining
(3) Expanding
(4) Reducing

97. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Predators in nature are 'prudent'.
Reason (R): If a predator is too efficient and overexploits its prey, then the prey might become extinct and following it, the predator will also become extinct for lack of food.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

98. Given below are two statements:
Statement I: In the *lac* operon, the repressor is synthesised from the *i* gene.
Statement II: The *lac* operon consists of four structural genes.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

99. With the help of given pedigree chart, find out the pattern of inheritance of the trait, given that among the progeny in F_3, females can only be carriers while all males are affected.



□: Unaffected male
■: Affected male
⦿: Carrier female
◯: Unaffected female
●: Affected female

(1) Y-linked
(2) X-linked recessive
(3) Autosomal dominant
(4) Autosomal recessive

100. The human chromosome that has 2968 genes, is;
(1) Chromosome Y.
(2) Chromosome X.
(3) Chromosome 1.
(4) Chromosome 21.

101. A structural feature of an ecosystem is;
(1) nutrient cycling.
(2) species composition.
(3) decomposition.
(4) energy flow.

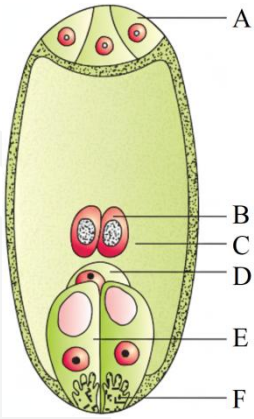
102. Match **List-I** with **List-II** w.r.t India.

List-I		List-II	
(A)	Biodiversity hotspots in India	(I)	14
(B)	National parks in India	(II)	3
(C)	Biosphere reserves	(III)	448
(D)	Wildlife sanctuaries	(IV)	90

Choose the *most appropriate* answer from the options given below.
(1) A-II, B-I, C-III, D-IV
(2) A-II, B-IV, C-I, D-III
(3) A-I, B-II, C-III, D-IV
(4) A-III, B-I, C-II, D-IV

103. Given below are two statements:
Statement I: Methanogens are present in the rumen of cattle and help in the breakdown of cellulose.
Statement II: The technology of biogas production was developed in India mainly due to the efforts of IARI and KVIC.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

104. Choose the **correct** statement for the labelled structures in the given diagram of a mature embryo sac.



(1) 'A' is located towards the micropylar end.
(2) 'F' is located towards the chalazal end.
(3) The ploidy of B, D and E is same.
(4) 'C' forms the zygote after double fertilisation.

105. At least one species is benefitted in the interaction between;
A. visiting flamingoes and resident fishes in some south American lakes.
B. dogs and ticks.
C. whale and barnacles.
D. cattle egret and foraging cattle.
E. tiger and deer.
Choose the **correct** answer from the options given below:
(1) A, B, C, D
(2) A, B, D only
(3) B, C, D, E
(4) B, D, E only

106. Identify the **incorrect** statement about Turner's syndrome.
(1) The affected individual is a male.
(2) Its pattern of inheritance cannot be studied by pedigree analysis.
(3) It is a case of aneuploidy.
(4) The affected individual is sterile.

107. Identify the **incorrect** statement about the organism $\phi \times 174$.
(1) It is a type of bacteriophage.
(2) In this virus, the flow of genetic information is in reverse direction.
(3) The length of its genetic material is defined as number of nucleotides in it.
(4) Its genetic material has 5386 nucleotides.

108. Match **List-I** with **List-II**.

List-I (Compound)		List-II (Source)	
(A)	Cyclosporin A	(I)	*Monascus purpureus*
(B)	Statins	(II)	*Streptococcus*
(C)	Streptokinase	(III)	*Trichoderma polysporum*
(D)	Ethanol	(IV)	*Saccharomyces cerevisiae*

Choose the *most appropriate* answer from the options given below.
(1) A-III, B-I, C-II, D-IV
(2) A-III, B-II, C-I, D-IV
(3) A-I, B-III, C-II, D-IV
(4) A-IV, B-I, C-II, D-III

109. Identify the **incorrectly** matched pair.
(1) False fruit - Cashew
(2) Parthenocarpic fruit - Banana
(3) True fruit - Strawberry
(4) Dry fruit - Groundnut

110. Identify the **correct** statement.
(1) Dodo is an example of recent extinction from Africa.
(2) Passenger pigeon became extinct due to alien species invasion.
(3) Nile Perch is an invasive weed in Amazon forests.
(4) The Amazon rain forest is being cut for cultivation of soyabeans or for conversion to grasslands for raising beef cattle.

111. Of the incident solar radiation, what percentage is photosynthetically active radiation (PAR)?
(1) Less than 50 per cent
(2) More than 50 per cent
(3) 100 per cent
(4) 90 per cent

112. Read the following statements regarding Mendel's experiments:
A. He conducted hybridisation experiments on garden peas for seven years.
B. He selected 14 pairs of true-breeding pea plant varieties.
C. Tall height is a recessive trait in garden pea.
D. He applied statistical analysis and mathematical logic to problems in biology.
E. The contrasting traits selected by him included inheritance of colour in flowers, pod and seed.
In the light of above statements choose the **correct** answer from the options given below:
(1) A, D and E are correct, B and C are incorrect
(2) A, B and E are correct, C and D are incorrect
(3) B, C and D are correct, A and E are incorrect
(4) A, C and D are correct, B and E are incorrect

113. Identify the **incorrectly** matched pair.
(1) AUG - Codes for Methionine
(2) UAA - Stop codon
(3) UGA - Terminator codon
(4) UAG - Initiator codon

114. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): If hybrids are made into apomicts, there is no segregation of characters in the hybrid progeny.
Reason (R): Apomixis is a form of sexual reproduction that mimics asexual reproduction.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

115. Many members of the genus _____ form mycorrhiza.
(1) *Trichoderma*
(2) *Aspergillus*
(3) *Glomus*
(4) *Lactobacillus*

116. Arrange the steps of DNA fingerprinting in the **correct** sequence:
A. Restriction digestion.
B. DNA isolation.
C. Autoradiography.
D. Electrophoresis.
E. DNA hybridisation by VNTR probe.
F. Southern blotting.
(1) B → A → D → F → E → C
(2) A → B → C → D → E → F
(3) C → F → E → A → B → D
(4) E → D → F → B → C → A

117. Read the following statements regarding sewage treatment:

A. Primary treatment involves physical removal of particles.
B. All solids that settle during primary treatment form the primary sludge.
C. In the aeration tanks, vigorous growth of anaerobic microbes takes place.
D. The effluent from the secondary treatment is generally released into natural water bodies like rivers and streams.
E. Flocs are formed in anaerobic digesters.

The **correct** statements are;
(1) A, B, C, E only
(2) A, B, D only
(3) B, C, D only
(4) A, C, D only

118. A child has blood group O. None of his parents had O blood group. What could be the possible genotype of the parents?
(1) *ii* and *ii*
(2) I^AI^A and I^BI^B
(3) I^Bi and I^Bi
(4) I^AI^A and I^Bi

119. The baculoviruses;
(1) are used as biofertilisers.
(2) are narrow spectrum species specific biocontrol agents.
(3) are not fit to be used in IPM programmes.
(4) have shown negative impacts on mammals.

120. Match **List-I** with **List-II**.

List-I (Population interaction)		**List-II (Representation)**	
(A)	Mutualism	(I)	-,0
(B)	Commensalism	(II)	+,+
(C)	Amensalism	(III)	+,0
(D)	Parasitism	(IV)	+,-

Choose the *most appropriate* answer from the options given below.
(1) A-II, B-III, C-I, D-IV
(2) A-II, B-I, C-III, D-IV
(3) A-III, B-II, C-I, D-IV
(4) A-IV, B-III, C-II, D-I

121. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The genotypic ratio at F_2 in the inheritance of flower colour in snapdragon deviated from Mendelian monohybrid cross.

Reason (R): The inheritance of flower colour in snapdragon is a case of incomplete dominance.

In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

122. Choose the **correct** sequence of structures as they are formed during embryo development in dicots.
(1) Proembryo → Globular embryo → Heart-shaped embryo → Mature embryo
(2) Proembryo → Heart-shaped embryo → Globular embryo → Mature embryo
(3) Globular embryo→ Heart-shaped embryo → Mature embryo → Proembryo
(4) Heart-shaped embryo → Globular embryo → Mature embryo → Proembryo

123. In a flowering plant, if the Megaspore Mother Cell (MMC) has 24 chromosomes, what will be the number of chromosomes in the nucellus cell and functional megaspore respectively?
(1) 24 and 12
(2) 24 and 24
(3) 12 and 12
(4) 12 and 24

124. Two closely related species competing for the same resources cannot co-exist indefinitely and the competitively inferior one will be eliminated eventually. This principle was stated by;
(1) Gause
(2) MacArthur
(3) Humboldt
(4) Ehrlich

125. Read the following statements regarding the structure of microsporangium and pollen grain.

A. The outer three wall layers of microsporangium perform the function of protection and help in dehiscence of anther to release the pollen.
B. Cells of the tapetum possess dense cytoplasm and generally have more than one nucleus.
C. The exine exhibits a fascinating array of patterns and designs.
D. The vegetative cell is small and floats in the cytoplasm of the generative cell.
E. Pollen grains of many species cause severe allergies and bronchial afflictions.

The **correct** statements are;
(1) A, B and C only
(2) A, B, C and E only
(3) B, C, D and E only
(4) A, C, D and E only

[Contd...

126. _____ used the frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes and 'mapped' their position on the chromosome.
(1) Thomas Hunt Morgan
(2) Alfred Sturtevant
(3) Henking
(4) Sutton and Boveri

127. The amount of available energy is maximum at;
(1) T_4 (2) T_1
(3) T_3 (4) T_2

128. Given below are two statements:
Statement I: Saprophytes have not been included in ecological pyramids.
Statement II: Pyramid of energy is always upright, can never be inverted.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

129. In bacteria, transcription and translation:
(1) occur in nucleus and cytoplasm, respectively.
(2) are catalysed by the same enzyme.
(3) involve processing of hnRNA and polypeptide formation, respectively.
(4) can be coupled.

130. The decomposers in a pond ecosystem include all the following, **except**;
(1) bacteria. (2) flagellates.
(3) fungi. (4) phytoplanktons.

131. Read the following statements regarding biodiversity patterns:
A. New York has more bird species than India.
B. Tropics harbour more species than polar areas.
C. The relation between species richness and area for a wide variety of taxa turns out to be a rectangular hyperbola.
D. Amazonian rain forest in South America has the greatest biodiversity on earth.
E. On a logarithmic scale, the species-area relationship is a straight line.
The number of **incorrect** statements is;
(1) four.
(2) three.
(3) one.
(4) two.

132. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Haemophilia	(I)	X-linked disorder
(B)	Sickle-cell anaemia	(II)	Found in the family members of Queen Victoria
(C)	Phenylketonuria	(III)	Classical example of point mutation
(D)	Colour blindness	(IV)	Pleiotropic autosomal recessive disorder

Choose the *most appropriate* answer from the options given below.
(1) A-II, B-III, C-I, D-IV
(2) A-II, B-III, C-IV, D-I
(3) A-III, B-II, C-IV, D-I
(4) A-I, B-IV, C-III, D-II

133. Which of the following is considered the most important cause driving animals and plants to extinction?
(1) Habitat loss and fragmentation
(2) Over-exploitation
(3) Alien species invasions
(4) Co-extinctions

134. Identify the **incorrect** statement about Francis Crick.
(1) He proposed the Central Dogma in molecular biology.
(2) He won Nobel prize in 1953.
(3) Along with Watson, he proposed the double helix model of DNA structure.
(4) He studied physics at University college, London.

135. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Autogamy	(I)	The only method of pollination in papaya flowers
(B)	Geitonogamy	(II)	One of the methods of pollination possible in chasmogamous flowers of *Viola*
(C)	Xenogamy	(III)	The only method of pollination in cleistogamous flowers

Choose the *most appropriate* answer from the options given below.
(1) A-I, B-II, C-III
(2) A-II, B-I, C-III
(3) A-III, B-I, C-II
(4) A-III, B-II, C-I

[Contd...

136. Which of the following constitutes the "seminal plasma"?
(1) Secretions of paired seminal vesicles and a prostate only
(2) Secretions of paired seminal vesicles, paired prostate glands and bulbourethral glands
(3) Sperms and secretions of the epididymis
(4) Secretions of paired seminal vesicles, a prostate, and paired bulbourethral glands

137. Which diagnostic method is commonly utilized for confirming cancer by examining tissue samples using histopathology?
(1) MRI scan (2) CT scan
(3) Biopsy (4) X ray

138. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Ti-plasmid	(I)	Used to deliver genes into animal cells
(B)	Retrovirus	(II)	Vector derived from *Agrobacterium tumefaciens*
(C)	pBR322	(III)	Virus that infects bacteria and used as vectors
(D)	Bacteriophage	(IV)	Specifically engineered *E.coli* cloning vector

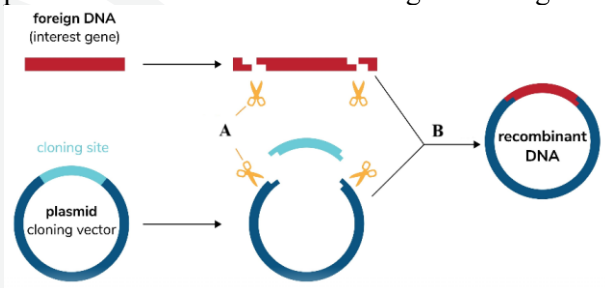
Choose the **correct** answer from the options given below:
(1) A-II, B-I, C-III, D-IV
(2) A-III, B-II, C-I, D-IV
(3) A-III, B-IV, C-I, D-II
(4) A-II, B-I, C-IV, D-III

139. Study the following examples of homology and analogy. Choose the set that represents Divergent Evolution:
A. Forelimbs of Cheetah and Human.
B. Eye of Octopus and Mammals.
C. Thorns of *Bougainvillea* and tendrils of *Cucurbita*.
D. Sweet potato and Potato.
E. Vertebrate hearts and brains.
(1) A, C, and E (2) B and D only
(3) A, B, and C (4) C, D, and E

140. Which of the following is an example of an "anamnestic response"?
(1) The low-intensity primary response to a new pathogen.
(2) The highly intensified secondary response to a previously encountered pathogen.
(3) The non-specific defense present at birth.
(4) The acceptance of a tissue graft by the body.

141. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Recombinant therapeutics do not induce unwanted immunological responses in humans.
Reason R: These therapeutics are non-identical to human proteins and are not isolated from non-human sources.
In the light of the above statements, choose the **correct** answer from the options given below.
(1) Both A and R are true, and R is the correct explanation of A.
(2) Both A and R are true, but R is not the correct explanation of A.
(3) A is true, but R is false.
(4) A is false, but R is true.

142. Which of the following enzymes catalyse the process A and B in the given diagram?



(1) A- DNA ligase, B- restriction endonuclease
(2) A- Taq DNA polymerase, B- Chitinase
(3) A- Restriction endonuclease, B- DNA polymerase
(4) A- Restriction endonuclease, B- DNA ligase

143. During the follicular phase of the menstrual cycle, the regeneration of the endometrium is primarily induced by:
(1) High levels of progesterone from the corpus luteum
(2) Secretion of estrogens by the growing follicles
(3) Low levels of LH and FSH
(4) Breakdown of the Graafian follicle

144. Regarding the menstrual cycle, identify the **correct** statements:
A. Menarche is the first menstruation that begins at puberty.
B. The LH surge induces the rupture of the Graafian follicle to release primary oocyte.
C. The corpus luteum secretes large amounts of estrogen to maintain the endometrium.
D. Lack of menstruation is always an indicator of pregnancy.
E. The luteal phase is also known as the secretory phase.
(1) A, B, and E only
(2) A, C, and D only
(3) A and E only
(4) A and B only

[Contd...

145. Transgenic mice are currently being developed and used specifically for:
(1) Testing the safety of polio vaccine.
(2) Production of Hepatitis B vaccine.
(3) Production of recombinant insulin.
(4) Production of human alpha-lactalbumin rich milk.

146. In the context of the Industrial Melanism observation in England, why did the count of melanic moths remain low in rural areas even after industrialization?
(1) Rural areas had higher populations of predators that preferred dark moths.
(2) Lichens, which are pollution indicators, continued to grow on trees in rural areas.
(3) White-winged moths in rural areas underwent saltation to become resistant to soot.
(4) Pollution occurred only in rural areas and not in urban areas.

147. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Golden rice	(I)	Genetically identical plants produced via tissue culture
(B)	Bt Cotton	(II)	Vitamin A enriched
(C)	Pomato	(III)	Pest resistance
(D)	Somaclones	(IV)	Somatic hybrid

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-I, D-IV
(2) A-III, B-II, C-I, D-IV
(3) A-III, B-IV, C-I, D-II
(4) A-II, B-III, C-IV, D-I

148. Which of the following pairs **correctly** matches a placental mammal with its corresponding Australian marsupial showing convergent evolution?
(1) Numbat — Spotted cuscus
(2) Anteater — Marsupial mouse
(3) Wolf — Tasmanian tiger cat
(4) Flying squirrel — Flying phalanger

149. Improved family planning programmes covering wider reproductive related areas are currently in operation under the popular name:
(1) RCH programmes.
(2) ART programmes.
(3) MTP programmes.
(4) test tube baby programmes.

150. Immediately after implantation, the inner cell mass (embryo) differentiates into an outer layer called _____ and an inner layer called _____.
(1) Ectoderm and Mesoderm
(2) Ectoderm and Endoderm
(3) Mesoderm and Endoderm
(4) Trophoblast and Chorion

151. Given below are two statements:
Statement I: The manipulation of living organisms by the human race cannot go on any further, without regulation.
Statement II: Genetically engineered insulin is made in bacterial cells yet it performs the same biological function as human insulin.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

152. Which of the following diseases do **not** spread by an insect vector?
(1) Ascariasis (2) Malaria
(3) Chikungunya (4) Dengue

153. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Acrosome	(I)	Collection of ovum
(B)	Mitochondria	(II)	Enzymes for fertilisation
(C)	Fimbriae	(III)	Secretion of milk
(D)	Cells of alveoli	(IV)	Energy source for sperm movement

Choose the **correct** answer from the options given below:
(1) A-II, B-IV, C-III, D-I
(2) A-II, B-IV, C-I, D-III
(3) A-III, B-IV, C-I, D-II
(4) A-II, B-I, C-IV, D-III

154. Which of the following regarding bioreactors is **incorrect**?
(1) They process large volumes (100–1000 litres) of culture.
(2) A stirred-tank reactor is usually cylindrical or with a flat base to facilitate mixing.
(3) They provide optimal growth conditions like temperature, pH, and oxygen.
(4) Sampling ports allow small volumes of culture to be withdrawn periodically

[Contd...

155. The method of directly injecting a sperm into ovum *in vitro* is done in a reproductive technology called;
(1) GIFT (2) ZIFT
(3) ICSI (4) ET

156. Complications due to various STIs may lead to:
(1) pelvic inflammatory diseases
(2) still births
(3) ectopic pregnancies
(4) all of these.

157. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Opioids	(I)	*Erythroxylum coca*
(B)	Cannabinoids	(II)	*Papaver somniferum*
(C)	Cocaine	(III)	*Datura*
(D)	Hallucinogens	(IV)	*Cannabis sativa*

Choose the **correct** answer from the options given below:
(1) A-II, B-IV, C-I, D-III
(2) A-III, B-II, C-I, D-IV
(3) A-III, B-IV, C-II, D-I
(4) A-II, B-IV, C-III, D-I

158. Which of the following statements regarding the European Federation of Biotechnology (EFB) definition of biotechnology is **correct**?
(1) It only focuses on traditional microbe-mediated processes like curd making.
(2) It encompasses both the traditional view and modern molecular biotechnology.
(3) It defines biotechnology as the use of only genetically modified organisms for large-scale production.
(4) It focuses exclusively on the integration of natural science for industrial-scale vaccines.

159. In 1983, which company successfully used *E. coli* to produce the A and B chains of human insulin?
(1) Pfizer
(2) Eli Lilly
(3) GlaxoSmithKline
(4) Merck

160. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: In gel electrophoresis, DNA fragments separate according to their size.
Reason R: Agarose gel provides a sieving effect that allows smaller fragments to move farther.
In the light of the above statements, choose the **correct** answer from the options given below.
(1) Both A and R are true, and R is the correct explanation of A.
(2) Both A and R are true, but R is not the correct explanation of A.
(3) A is true, but R is false.
(4) A is false, but R is true.

161. Which of the following is a hormone releasing IUD?
(1) Multiload 375 (2) Cu7
(3) LNG-20 (4) CuT

162. What is the approximate percentage of lymphoid tissue represented by Mucosa-associated lymphoid tissue (MALT) in the human body?
(1) 10 per cent (2) 25 per cent
(3) 50 per cent (4) 75 per cent

163. Regarding RNA interference (RNAi), which of the following statements are **correct**?
A. It takes place in all eukaryotic organisms as a method of cellular defense.
B. It involves the use of retrovirus vectors to introduce nematode-specific genes.
C. It results in the silencing of specific mRNA.
D. The source of complementary RNA is always from a DNA virus.
E. It produces only sense RNA in the host cells.
(1) A, B, and C only
(2) B, C, and D only
(3) C, D, and E only
(4) A and C only

164. Which of the following statements is **incorrect** about the procedure of vasectomy?
(1) The oviducts are cut and tied.
(2) It blocks gamete transport.
(3) It is a surgical method of birth control in males.
(4) It is an irreversible method.

165. Match **List-I** with **List-II**.

List-I (Embryo development stage)		List-II (Timelines)	
(A)	Heart formation	(I)	End of 24 weeks
(B)	Limbs and digits	(II)	After 1 month
(C)	Major organ systems	(III)	End of 2nd month
(D)	Eyelashes formed	(IV)	End of 12 weeks

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-I, D-IV
(2) A-III, B-II, C-I, D-IV
(3) A-III, B-IV, C-I, D-II
(4) A-II, B-III, C-IV, D-I

[Contd...

166. Smoking reduces the concentration of haembound oxygen in the blood by increasing the content of:
(1) Carbon dioxide (CO_2)
(2) Carbon monoxide (CO)
(3) Nitrogen
(4) Nicotine

167. Which of the following is a **correct** statement regarding the Polymerase Chain Reaction (PCR)?
(1) It is used to amplify proteins using a heat-labile DNA polymerase.
(2) Each PCR cycle consists of three sequential steps: Denaturation then Extension, and finally Annealing.
(3) *Taq* polymerase, isolated from a bacterium, remains active during high-temperature denaturation.
(4) Primers used in PCR are double-stranded long DNA sequences complementary to the target region.

168. All of the following are the use of amniocentesis, but one is misuse. Which one is misuse?
(1) Detection of Down syndrome in the developing fetus.
(2) Detection of hemophilia or other genetic linkages.
(3) Diagnosis of disorders like sickle cell anaemia.
(4) Determination of the sex of the unborn child.

169. Match **List-I** with **List-II**.

List-I		List-II	
(A)	First cellular forms of life	(I)	350 mya
(B)	Invertebrates formed	(II)	2000 mya
(C)	Jawless fish evolved	(III)	500 mya
(D)	Sea weeds and few plants	(IV)	320 mya

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-I, D-IV
(2) A-III, B-II, C-I, D-IV
(3) A-III, B-IV, C-I, D-II
(4) A-III, B-II, C-IV, D-I

170. For normal fertility in human males, at least what percentage of sperms should have normal shape and size?
(1) 30%
(2) 40%
(3) 60%
(4) 80%

171. Which of the following shows embryological evidence of evolution?
(1) Fossils in different sedimentary layers.
(2) Vestigial gill slits in embryos of all vertebrates.
(3) Forelimbs of mammals.
(4) Potato and sweet potato.

172. Consider the following statements regarding the life cycle of *Plasmodium*:
A. The infectious stage for humans is the sporozoite, which is stored in the mosquito's salivary glands.
B. Parasites first multiply sexually within the liver cells before attacking RBCs.
C. The rupture of RBCs releases haemozoin, which causes recurring chills and high fever.
D. Fertilization and development of the parasite take place in the human blood stream.
Choose the **correct** statements:
(1) A and C only
(2) A, B, and D only
(3) B, C, and D only
(4) A, B, C and D

173. In Miller's experiment, the conditions created in the closed flask to simulate the primitive earth atmosphere were:
(1) CH_4, H_2, NH_3 and water vapor at 800 °C with electric discharge.
(2) CH_4, H_2, NO_2 and water vapor at 800 °C with UV rays.
(3) O_2, CH_4, H_2 and NH_3 at 600 °C with electric discharge.
(4) NH_3, CH_4, H_2 and liquid water at 100 °C with catalytic converters.

174. Match **List-I** with **List-II**.

List-I		List-II	
(A)	GEAC	(I)	Unauthorized use of bio-resources
(B)	Biopiracy	(II)	2nd amendment of Indian Patents Bill
(C)	Basmati Patent by an American company	(III)	Validity of GM research
(D)	Patent terms of emergency provisions	(IV)	US Patent and Trademark Office

Choose the **correct** answer from the options given below:
(1) A-III, B-I, C-II, D-IV
(2) A-III, B-I, C-IV, D-II
(3) A-III, B-IV, C-I, D-II
(4) A-III, B-II, C-IV, D-I

[Contd...

175. Which of the following protein is used to treat the medical condition Emphysema?
(1) Alpha-lactalbumin
(2) Adenosine deaminase
(3) Alpha-1-antitrypsin
(4) *Cry*IAb protein

176. The cavity of the tertiary follicle is called:
(1) Antrum
(2) Blastocoel
(3) Perivitelline space
(4) Amniotic cavity

177. In a population at Hardy-Weinberg equilibrium, if the frequency of a recessive allele (a) is 0.4, what is the frequency of the heterozygous (Aa) individuals?
(1) 0.16 (2) 0.36
(3) 0.48 (4) 0.60

178. Choose the **correct** statement about the first recombinant DNA molecule:
(1) It was constructed by linking a gene encoding ampicillin resistance with a plasmid of *E. coli*.
(2) Stanley Cohen and Herbert Boyer accomplished this in 1963.
(3) It involved the native plasmid of *Salmonella typhimurium*.
(4) The recombinant DNA was transferred into *Salmonella* to replicate using its DNA polymerase.

179. In severe cases of pneumonia, which of the following characteristic physical signs may appear?
(1) Nasal congestion and discharge.
(2) Dry, scaly lesions on the skin.
(3) Lips and finger nails turning gray to bluish in colour.
(4) Chronic inflammation of the lower limbs.

180. Given below are two statements:
Statement I: Origin of replication is a sequence where translation of the linked DNA starts.
Statement II: Resistance genes for antibiotics like kanamycin are useful selectable markers.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

■■■

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